

Supplementary Table 9. Reviewer-requested sensitivity analyses for development-only model selection, SSL PCA dimensionality reduction, elastic-net regularization and pH-threshold outcome definitions

Frontiers in Signal Processing R1 supplementary table

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**Supplementary Table 9. Reviewer-requested sensitivity analyses for development-only model selection, SSL PCA dimensionality reduction, elastic-net regularization and pH-threshold outcome definitions**

Standalone display version for LaTeX upload/review. Machine-readable source: `source_tables/Supplementary_Table_9_reviewer_sensitivity_analyses.csv`. Rows: 24; columns: 24. The source table is split below into source-defined sections and column blocks so that all non-empty cells remain readable.

**analysis type: Development model selection**

Table 1: Supplementary Table 9, Development model selection, column block 1  
of 3

| config                  | model                    | selection split                                                                    | Development train records | Development validation records |
|-------------------------|--------------------------|------------------------------------------------------------------------------------|---------------------------|--------------------------------|
| p80 d64 l3 channel cw02 | Signal + SSL             | 75% original-training records fit /<br>25% original-training records<br>validation | 289                       | 97                             |
| p80 d64 l3 channel cw02 | Signal + SSL + phenotype | 75% original-training records fit /<br>25% original-training records<br>validation | 289                       | 97                             |
| p60 d64 l2 mixed cw02   | Signal + SSL             | 75% original-training records fit /<br>25% original-training records<br>validation | 289                       | 97                             |
| p60 d64 l2 mixed cw02   | Signal + SSL + phenotype | 75% original-training records fit /<br>25% original-training records<br>validation | 289                       | 97                             |
| p80 d48 l2 mixed cw01   | Signal + SSL             | 75% original-training records fit /<br>25% original-training records<br>validation | 289                       | 97                             |
| p80 d48 l2 mixed cw01   | Signal + SSL + phenotype | 75% original-training records fit /<br>25% original-training records<br>validation | 289                       | 97                             |
| p120 d64 l2 block cw02  | Signal + SSL             | 75% original-training records fit /<br>25% original-training records<br>validation | 289                       | 97                             |
| p120 d64 l2 block cw02  | Signal + SSL + phenotype | 75% original-training records fit /<br>25% original-training records<br>validation | 289                       | 97                             |

Table 2: Supplementary Table 9, Development model selection, column block 2  
of 3

| config                  | model                    | Development train events | Development validation events | AUROC  |
|-------------------------|--------------------------|--------------------------|-------------------------------|--------|
| p80 d64 l3 channel cw02 | Signal + SSL             | 59                       | 20                            | 0.7078 |
| p80 d64 l3 channel cw02 | Signal + SSL + phenotype | 59                       | 20                            | 0.7045 |
| p60 d64 l2 mixed cw02   | Signal + SSL             | 59                       | 20                            | 0.6942 |
| p60 d64 l2 mixed cw02   | Signal + SSL + phenotype | 59                       | 20                            | 0.6981 |
| p80 d48 l2 mixed cw01   | Signal + SSL             | 59                       | 20                            | 0.7396 |
| p80 d48 l2 mixed cw01   | Signal + SSL + phenotype | 59                       | 20                            | 0.75   |
| p120 d64 l2 block cw02  | Signal + SSL             | 59                       | 20                            | 0.6519 |
| p120 d64 l2 block cw02  | Signal + SSL + phenotype | 59                       | 20                            | 0.6597 |

Table 3: Supplementary Table 9, Development model selection, column block 3  
of 3

| config                  | model                    | AUPRC  | Brier score | event prevalence |
|-------------------------|--------------------------|--------|-------------|------------------|
| p80 d64 l3 channel cw02 | Signal + SSL             | 0.3569 | 0.2411      | 0.2062           |
| p80 d64 l3 channel cw02 | Signal + SSL + phenotype | 0.3569 | 0.2414      | 0.2062           |
| p60 d64 l2 mixed cw02   | Signal + SSL             | 0.3415 | 0.2282      | 0.2062           |
| p60 d64 l2 mixed cw02   | Signal + SSL + phenotype | 0.3564 | 0.2287      | 0.2062           |
| p80 d48 l2 mixed cw01   | Signal + SSL             | 0.4291 | 0.2277      | 0.2062           |
| p80 d48 l2 mixed cw01   | Signal + SSL + phenotype | 0.4482 | 0.2267      | 0.2062           |
| p120 d64 l2 block cw02  | Signal + SSL             | 0.2925 | 0.2495      | 0.2062           |
| p120 d64 l2 block cw02  | Signal + SSL + phenotype | 0.3    | 0.2467      | 0.2062           |

analysis type: development selected test evaluation

Table 4: Supplementary Table 9, development selected test evaluation, column  
block 1 of 3

| config                | model                    | selection split                                                                                | AUROC  | AUPRC  |
|-----------------------|--------------------------|------------------------------------------------------------------------------------------------|--------|--------|
| p80 d48 l2 mixed cw01 | Signal + SSL + phenotype | selected by original-training internal<br>validation; original held-out test<br>evaluated once | 0.6348 | 0.3872 |

Table 5: Supplementary Table 9, development selected test evaluation, column  
block 2 of 3

| config                | model                    | Brier score | event prevalence | Train records |
|-----------------------|--------------------------|-------------|------------------|---------------|
| p80 d48 l2 mixed cw01 | Signal + SSL + phenotype | 0.2542      | 0.2048           | 386           |

Table 6: Supplementary Table 9, development selected test evaluation, column  
block 3 of 3

| config                | model                    | Held-out test records | Train events | Held-out test events |
|-----------------------|--------------------------|-----------------------|--------------|----------------------|
| p80 d48 l2 mixed cw01 | Signal + SSL + phenotype | 166                   | 79           | 34                   |



analysis type: dimensionality regularization

Table 7: Supplementary Table 9, dimensionality regularization, column block 1 of 4

| model                                            | AUROC  | AUPRC  | Brier score | event prevalence |
|--------------------------------------------------|--------|--------|-------------|------------------|
| signal + ssl64 + signal phenotype l2             | 0.6168 | 0.4016 | 0.2652      | 0.2048           |
| Signal + SSL pca5 + signal phenotype l2          | 0.6286 | 0.3863 | 0.2547      | 0.2048           |
| Signal + SSL pca5 + signal phenotype elasticnet  | 0.6279 | 0.3801 | 0.2548      | 0.2048           |
| Signal + SSL pca10 + signal phenotype l2         | 0.6212 | 0.4241 | 0.2588      | 0.2048           |
| Signal + SSL pca10 + signal phenotype elasticnet | 0.6194 | 0.4026 | 0.2584      | 0.2048           |
| Signal + SSL pca20 + signal phenotype l2         | 0.6535 | 0.4256 | 0.2566      | 0.2048           |
| Signal + SSL pca20 + signal phenotype elasticnet | 0.6549 | 0.4172 | 0.2557      | 0.2048           |

Table 8: Supplementary Table 9, dimensionality regularization, column block 2  
of 4

| model                                               | AUROC  | Train records | Held-out test records | Train events |
|-----------------------------------------------------|--------|---------------|-----------------------|--------------|
| signal + ssl64 + signal phenotype l2                | 0.6168 | 386           | 166                   | 79           |
| Signal + SSL pca5 + signal<br>phenotype l2          | 0.6286 | 386           | 166                   | 79           |
| Signal + SSL pca5 + signal<br>phenotype elasticnet  | 0.6279 | 386           | 166                   | 79           |
| Signal + SSL pca10 + signal<br>phenotype l2         | 0.6212 | 386           | 166                   | 79           |
| Signal + SSL pca10 + signal<br>phenotype elasticnet | 0.6194 | 386           | 166                   | 79           |
| Signal + SSL pca20 + signal<br>phenotype l2         | 0.6535 | 386           | 166                   | 79           |
| Signal + SSL pca20 + signal<br>phenotype elasticnet | 0.6549 | 386           | 166                   | 79           |

Table 9: Supplementary Table 9, dimensionality regularization, column block 3 of 4

| model                                            | AUROC  | Held-out test events | SSL dimensions | penalty    |
|--------------------------------------------------|--------|----------------------|----------------|------------|
| signal + ssl64 + signal phenotype l2             | 0.6168 | 34                   | 64             | l2         |
| Signal + SSL pca5 + signal phenotype l2          | 0.6286 | 34                   | 5              | l2         |
| Signal + SSL pca5 + signal phenotype elasticnet  | 0.6279 | 34                   | 5              | elasticnet |
| Signal + SSL pca10 + signal phenotype l2         | 0.6212 | 34                   | 10             | l2         |
| Signal + SSL pca10 + signal phenotype elasticnet | 0.6194 | 34                   | 10             | elasticnet |
| Signal + SSL pca20 + signal phenotype l2         | 0.6535 | 34                   | 20             | l2         |
| Signal + SSL pca20 + signal phenotype elasticnet | 0.6549 | 34                   | 20             | elasticnet |

Table 10: Supplementary Table 9, dimensionality regularization, column block  
4 of 4

| model                                            | AUROC  | n predictors before one hot | events per predictor before one hot | PCA explained variance ratio |
|--------------------------------------------------|--------|-----------------------------|-------------------------------------|------------------------------|
| signal + ssl64 + signal phenotype l2             | 0.6168 | 77                          | 1.026                               |                              |
| Signal + SSL pca5 + signal phenotype l2          | 0.6286 | 18                          | 4.389                               | 0.84                         |
| Signal + SSL pca5 + signal phenotype elasticnet  | 0.6279 | 18                          | 4.389                               | 0.84                         |
| Signal + SSL pca10 + signal phenotype l2         | 0.6212 | 23                          | 3.435                               | 0.9567                       |
| Signal + SSL pca10 + signal phenotype elasticnet | 0.6194 | 23                          | 3.435                               | 0.9567                       |
| Signal + SSL pca20 + signal phenotype l2         | 0.6535 | 33                          | 2.394                               | 0.9953                       |
| Signal + SSL pca20 + signal phenotype elasticnet | 0.6549 | 33                          | 2.394                               | 0.9953                       |

analysis type: outcome sensitivity

Table 11: Supplementary Table 9, outcome sensitivity, column block 1 of 4

| model                     | AUROC  | AUPRC  | Brier score | event prevalence |
|---------------------------|--------|--------|-------------|------------------|
| Classical signal features | 0.6303 | 0.3366 | 0.2525      | 0.2048           |
| Signal + SSL + phenotype  | 0.6168 | 0.4016 | 0.2652      | 0.2048           |
| Classical signal features | 0.6398 | 0.3882 | 0.2525      | 0.1988           |
| Signal + SSL + phenotype  | 0.6744 | 0.4432 | 0.2489      | 0.1988           |
| Classical signal features | 0.7809 | 0.4773 | 0.2183      | 0.1145           |
| Signal + SSL + phenotype  | 0.8006 | 0.487  | 0.1941      | 0.1145           |
| Classical signal features | 0.8181 | 0.5409 | 0.207       | 0.0904           |
| Signal + SSL + phenotype  | 0.8631 | 0.526  | 0.1726      | 0.0904           |

Table 12: Supplementary Table 9, outcome sensitivity, column block 2 of 4

| model                     | AUROC  | Train records | Held-out test records | Train events |
|---------------------------|--------|---------------|-----------------------|--------------|
| Classical signal features | 0.6303 | 386           | 166                   | 79           |
| Signal + SSL + phenotype  | 0.6168 | 386           | 166                   | 79           |
| Classical signal features | 0.6398 | 386           | 166                   | 72           |
| Signal + SSL + phenotype  | 0.6744 | 386           | 166                   | 72           |
| Classical signal features | 0.7809 | 386           | 166                   | 37           |
| Signal + SSL + phenotype  | 0.8006 | 386           | 166                   | 37           |
| Classical signal features | 0.8181 | 386           | 166                   | 25           |
| Signal + SSL + phenotype  | 0.8631 | 386           | 166                   | 25           |



Table 13: Supplementary Table 9, outcome sensitivity, column block 3 of 4

| model                     | AUROC  | Held-out test events | outcome         | outcome definition                                                |
|---------------------------|--------|----------------------|-----------------|-------------------------------------------------------------------|
| Classical signal features | 0.6303 | 34                   | neonatal risk   | primary composite: cord pH $\leq 7.15$ or 5-minute Apgar $\leq 7$ |
| Signal + SSL + phenotype  | 0.6168 | 34                   | neonatal risk   | primary composite: cord pH $\leq 7.15$ or 5-minute Apgar $\leq 7$ |
| Classical signal features | 0.6398 | 33                   | cord ph lt 7 15 | cord pH $\leq 7.15$ only                                          |
| Signal + SSL + phenotype  | 0.6744 | 33                   | cord ph lt 7 15 | cord pH $\leq 7.15$ only                                          |
| Classical signal features | 0.7809 | 19                   | cord ph lt 7 10 | cord pH $\leq 7.10$ only                                          |
| Signal + SSL + phenotype  | 0.8006 | 19                   | cord ph lt 7 10 | cord pH $\leq 7.10$ only                                          |
| Classical signal features | 0.8181 | 15                   | cord ph lt 7 05 | cord pH $\leq 7.05$ only                                          |
| Signal + SSL + phenotype  | 0.8631 | 15                   | cord ph lt 7 05 | cord pH $\leq 7.05$ only                                          |

Table 14: Supplementary Table 9, outcome sensitivity, column block 4 of 4

| model                     | AUROC  | AUPRC / test prevalence |
|---------------------------|--------|-------------------------|
| Classical signal features | 0.6303 | 1.643                   |
| Signal + SSL + phenotype  | 0.6168 | 1.961                   |
| Classical signal features | 0.6398 | 1.953                   |
| Signal + SSL + phenotype  | 0.6744 | 2.229                   |
| Classical signal features | 0.7809 | 4.17                    |
| Signal + SSL + phenotype  | 0.8006 | 4.255                   |
| Classical signal features | 0.8181 | 5.985                   |
| Signal + SSL + phenotype  | 0.8631 | 5.821                   |